

Xinru Wang

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 [Google Scholar](#)

Summary My research intersects Human-Computer Interaction (HCI) and AI, with a focus on Explainable AI (XAI). I iterate on the **evaluation and design of user-centered XAI systems for AI-assisted human decision-making**. My research considers the contextual relevance in fast evolving *AI models* and diverse *user interfaces*, with an emphasis on making AI explanations *collaborative, adaptable, and glanceable*.

Education **Purdue University** West Lafayette, IN
Ph.D. in Computer Science Aug. 2019 – Aug. 2025
Thesis: Human-Centered Evaluation and Design of AI Explanations in AI-Assisted Decision-Making
Thesis Committee: [Prof. Ming Yin](#) (Advisor), Prof. Ninghui Li, Prof. Alexandros Psomas, Prof. Tianyi Zhang

Peking University Beijing, China
B.S. in Psychology (Dual Degree) Sept. 2016 – Jun. 2019
B.S. in Intelligence Science and Technology Sept. 2015 – Jun. 2019
Thesis: “Analysis of MOOC Forum Data towards AI Support”
Thesis Advisor: Prof. Xiaoming Li

Publications **Beyond Recommender: An Exploratory Study of the Effects of Different AI Roles in AI-Assisted Decision Making**
Shuai Ma, Chenyi Zhang, **Xinru Wang**, Xiaojuan Ma, Ming Yin.
Under Review

XAgent: An Explainability Tool for Identifying and Correcting Failures in Multi-Agent Workflows
Xinru Wang, Ming Yin, Eunye Koh, Mustafa Doga Dogan.
Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA). 2026

Towards Human-AI Deliberation: Design and Evaluation of LLM-Empowered Deliberative AI for AI-Assisted Decision-Making
Shuai Ma, Qiaoyi Chen, **Xinru Wang**, Chengbo Zheng, Zhenhui Peng, Ming Yin, Xiaojuan Ma.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2025.

Less or More: Towards Glanceable Explanations for LLM Recommendations Using Ultra-Small Devices

Xinru Wang, Mengjie Yu, Hannah Nguyen, Michael Iuzzolino, Tianyi Wang, Peiqi Tang, Natasha Lynova, Co Tran, Ting Zhang, Naveen Sendhilnathan, Hrvoje Benko, Haijun Xia, Tanya Jonker.

Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI). 2025.

Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels

Xinru Wang, Hannah Kim, Sajjadur Rahman, Kushan Mitra, Zhengjie Miao.

Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2024.

Human-Centered Evaluation of Explanations in AI-Assisted Decision-Making

Xinru Wang.

Companion Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI Doctoral Consortium). 2024.

“Are You Really Sure?” Understanding the Effects of Human Self-Confidence Calibration in AI-Assisted Decision Making

Shuai Ma, **Xinru Wang**, Ying Lei, Chuhan Shi, Ming Yin, Xiaojuan Ma.

Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2024.

The Effects of AI Biases and Explanations on Human Decision Fairness: A Case Study of Bidding in Rental Housing Markets

Xinru Wang, Chen Liang, Ming Yin.

Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI). 2023.

Who Should I Trust: AI or Myself? Leveraging Human and AI Correctness Likelihood to Promote Appropriate Trust in AI-Assisted Decision-Making

Shuai Ma, Ying Lei, **Xinru Wang**, Chengbo Zheng, Chuhan Shi, Ming Yin, Xiaojuan Ma.

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI). 2023.

Watch Out For Updates: Understanding the Effects of Model Explanation Updates in AI-Assisted Decision Making

Xinru Wang, Ming Yin.

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI). 2023.

Will You Accept the AI Recommendation? Predicting Human Behavior in AI-Assisted Decision Making

Xinru Wang, Zhuoran Lu, Ming Yin.

Proceedings of the 2022 ACM Web Conference (WWW). 2022.

Effects of Explanations in AI-Assisted Decision Making: Principles and Comparisons

Xinru Wang, Ming Yin.

ACM Transactions on Interactive Intelligent Systems (TiiS). 2022

Are Explanations Helpful? A Comparative Study of the Effects of Explanations in AI-Assisted Decision-Making

Xinru Wang, Ming Yin.

Proceedings of the 26th International Conference on Intelligent User Interfaces (IUI). 2021.

Professional Experience	Singapore-MIT Alliance for Research and Technology	Singapore
	Research Intern	Sep. 2025 - Present
	– Advisor: Prof. Thomas Malone	
	Adobe Research	San Jose, CA, US
	Research Intern	May. 2025 - Aug. 2025
	– Mentor: Doga Dogan, Eunye Koh	
	– Build an explainable interface to supports users with varying AI expertise in identifying and correcting failures in multi-agent workflows. Research outcomes facilitated collaboration between Adobe developers and domain experts on the AEM Sites Optimizer.	
	Purdue University	West Lafayette, IN, US
	Research Assistant	Aug. 2019 - Aug. 2025
	– Advisor: Prof. Ming Yin	
	– Conducted large-scale online user studies to evaluate the impact of AI explanation designs on human decision-making processes.	
	Meta Reality Labs	Redmond, WA, US
	Research Intern	Sep. 2023 - Mar. 2024
	– Mentor: Anna Mengjie Yu, Tanya Jonker	
	– Developed methods for delivering glanceable explanations for LLM-driven personal assistants on ultra-small devices. Research insights contributed to the development of explainable AI features on Meta’s future AR devices.	
	Megagon Labs	Mountain View, CA, US
	Research Intern	May. 2023 - Aug. 2023

– *Mentor: Hannah Kim, Zhengjie Miao*

– Developed and evaluated a human-LLM collaborative annotation framework for natural language tasks. The findings directly contributed to the development of MEGAnno+, the company’s newly released data annotation tool.

University of Michigan

Ann Arbor, MI, US

Summer Research Intern

Jul. 2018 - Sep. 2018

– *Advisor: Prof. Lionel Robert*

– Investigated bi-directional trust in semi-autonomous vehicles through user testing and trust modeling.

Kendall Square Capital

Beijing, China

Machine Learning Intern

Jan. 2019 - Apr. 2019

– Extracted time-series features from the limit order book data for prediction of mid-price movement in high frequency trading.

DiDi

Beijing, China

Machine Learning Intern

Sep. 2018 - Jan. 2019

– Implemented machine learning models to predict drivers’ order-taken behavior and waiting-time on a million-scale database.

Selected
Talks

User-Centered Evaluation and Design of AI Explanations in AI-Assisted Decision-Making

Presented at UNC Chapel Hill

Feb. 2025

Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels

Presented at CHI 2024

May. 2024

Human-Centered Evaluation of Explanations in AI-Assisted Decision-Making

Presented at IUI 2024 Doctoral Consortium

Apr. 2024

Will You Accept the AI Recommendation? Predicting Human Behavior in AI-Assisted Decision Making

Presented at WWW 2022

Apr. 2022

Understanding Human Behavior in AI-Assisted Decision Making: Experiments and Models

Presented at [MIDAS Future Leaders Summit 2022](#)

Apr. 2022

Are Explanations Helpful? A Comparative Study of the Effects of Explanations in AI-Assisted Decision-Making

Presented at IUI 2021

Apr. 2021

Service

Program Committee

ACM Conference on Designing Interactive Systems (DIS): 2026
ACM Conference on Human Factors in Computing Systems (CHI): 2026
ACM Conference on Fairness, Accountability, and Transparency (FAccT): 2025, 2023
ACM Symposium on Eye Tracking Research & Applications (ETRA): 2025
ACM Conference on Intelligent User Interfaces (IUI): 2025
ACM CHI Workshop on Human-Centered Explainable AI (HCXAI): 2024, 2023, 2022

Assistant to Subcommittee Chair

ACM Conference on Human Factors in Computing Systems (CHI): 2025

Conference Reviewer

ACM Conference on Human Factors in Computing Systems (CHI) Poster: 2026
ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW): 2026
ACM Symposium on User Interface Software and Technology (UIST): 2025
ACM Conference on Human Factors in Computing Systems (CHI): 2025, 2024, 2023
ACM Conference on Human Factors in Computing Systems (CHI) Late-Breaking Work: 2025, 2024, 2023
ACM Conference on Designing Interactive Systems (DIS): 2024
ACM/IEEE International Conference on Human-Robot Interaction (HRI) alt.HRI: 2024

Journal Reviewer

Transactions on Computer-Human Interaction (TOCHI)
IEEE Transactions on Human-Machine Systems (THMS)
ACM Transactions on Interactive Intelligent Systems (TiiS)
Transactions on Intelligent Systems and Technology (TIST)

Student Volunteer

ACM Conference on Intelligent User Interfaces (IUI): 2024
ACM Conference on Research and Development in Information Retrieval (SIGIR): 2018

Teaching

CS475 Human-Computer Interaction, Purdue University

Graduate Teaching Assistant for Prof. Ming Yin

Spring 2025

- Undergraduate-level course with 90 students.
- Developed homework questions and graded assignments.

CS471 Introduction to Artificial Intelligence, Purdue University

	Head Graduate Teaching Assistant for Prof. Raymond A. Yeh	Spring 2023
	– Undergraduate-level course with 200 students.	
	– Led a team of five Teaching Assistants, developed homework questions, graded assignments, and conducted office hours.	
	– Recognized with the Graduate Teaching Award in the CS Department.	
	Head Graduate Teaching Assistant for Prof. Ming Yin	Summer 2022, Fall 2022
	Graduate Teaching Assistant for Prof. Jennifer Neville	Fall 2019
	CS242 Introduction To Data Science , Purdue University	
	Graduate Teaching Assistant for Prof. Tony Bergstrom	Spring 2020
	– Undergraduate-level course with 150 students.	
	– Developed homework questions.	
	CS38001 Python Programming , Purdue University	
	Graduate Teaching Assistant for Prof. Ruby Tahboub	Fall 2019, Spring 2020
	– Undergraduate-level course with 70 students.	
	– Developed homework questions, graded assignments, and led lab sessions and office hours.	
Mentoring	Songyu He (HKUST undergraduate student)	Fall 2024
	Hannah Nguyen (Meta Reality Labs research assistant)	Spring 2024
Honors and Awards	NSF Student Travel Award (IUI 2024)	2024
	Apple PhD Fellowship Nominee (Purdue University)	2024
	Graduate Teaching Award (Purdue University)	2023
	Summit Attendee and Speaker (MIDAS Future Leaders Summit)	2022
	Women in Science Program Travel Grant (Purdue University)	2022
	Microsoft Research PhD Fellowship Nominee (Purdue University)	2022
	Academic Excellence Award (Peking University, top 10%)	2018
	Fei-Xun Scholarship (Peking University)	2018
	Academic Excellence Award (Peking University, top 15%)	2016
	May 4th Scholarship (Peking University)	2016
Skills	Programming: Python, R, HTML/CSS/JavaScript, SQL, MATLAB, C/C++, Java Tools & Frameworks: Flask, Pytorch, Sklearn, OpenAI, HuggingFace, Meteor User Study: Online (AMTurk, Prolific) and in-lab Data Analysis: Quantitative (e.g. ANOVA, regressions, SEM) and qualitative (e.g. thematic analysis) Languages: English (GRE 162+170+3.5, TOEFL 111), Mandarin (Native)	